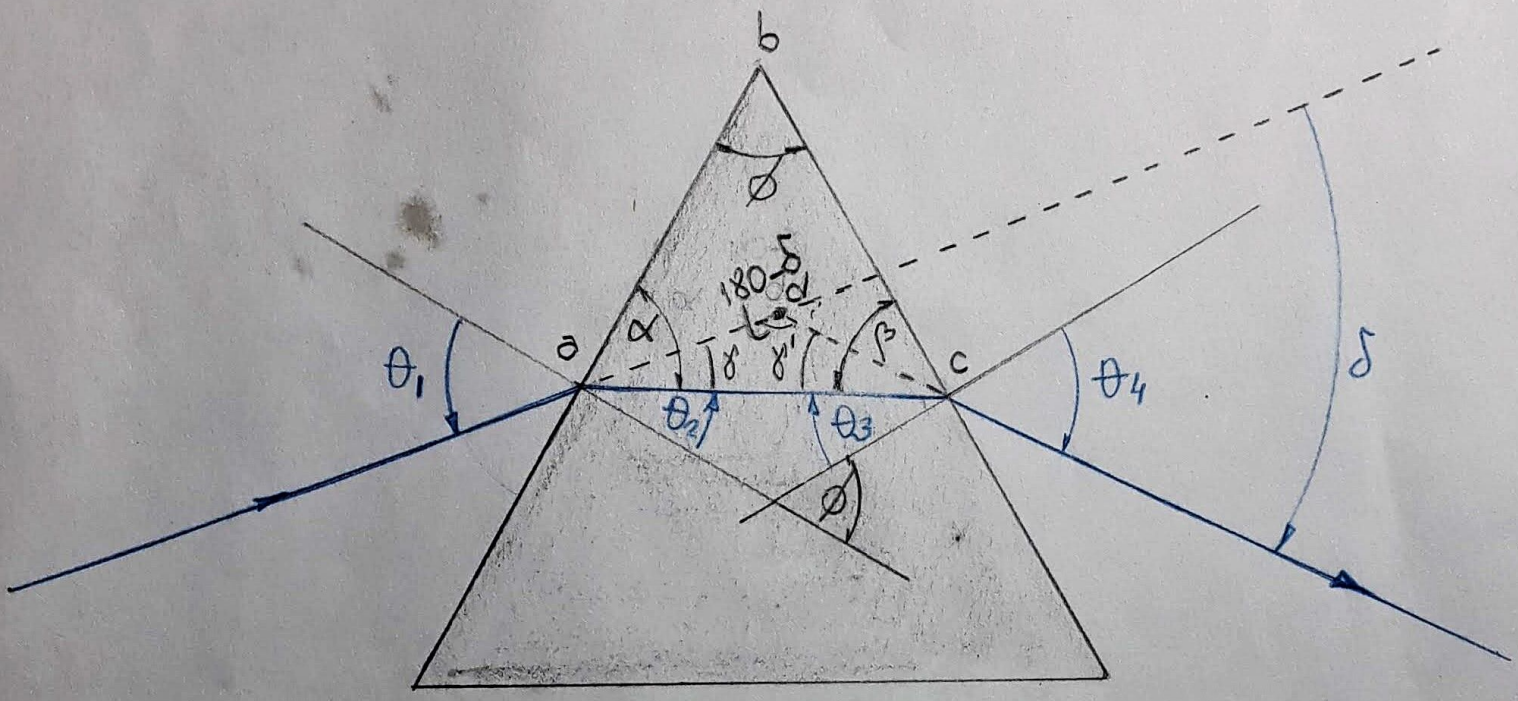
Repaso del teórico



$\hat{abc}$: $\alpha + \beta + \phi = 180°$

$90° - \theta_2 + 90° - \theta_3 + \phi = 180°$

$\boxed{\theta_3 = \phi - \theta_2}$

$\alpha + \theta_2 = 90°$ ^ $\beta + \theta_3 = 90°$

$\phi = \theta_2 + \theta_3$

n_{aire} sen θ_1 = n_{vidrio} sen θ_2

$$\boxed{\theta_2 = \text{arc sen}\left(\frac{\text{sen}\,\theta_1}{n_{vidrio}}\right)}$$

n_{vidrio} sen θ_3 = n_{aire} sen θ_4

$$\boxed{\theta_4 = \text{arc sen}\left(n_{vidrio}\ \text{sen}\,\theta_3\right)}$$

$\hat{adc}$: $\gamma + \gamma' + 180° - \delta = 180°$

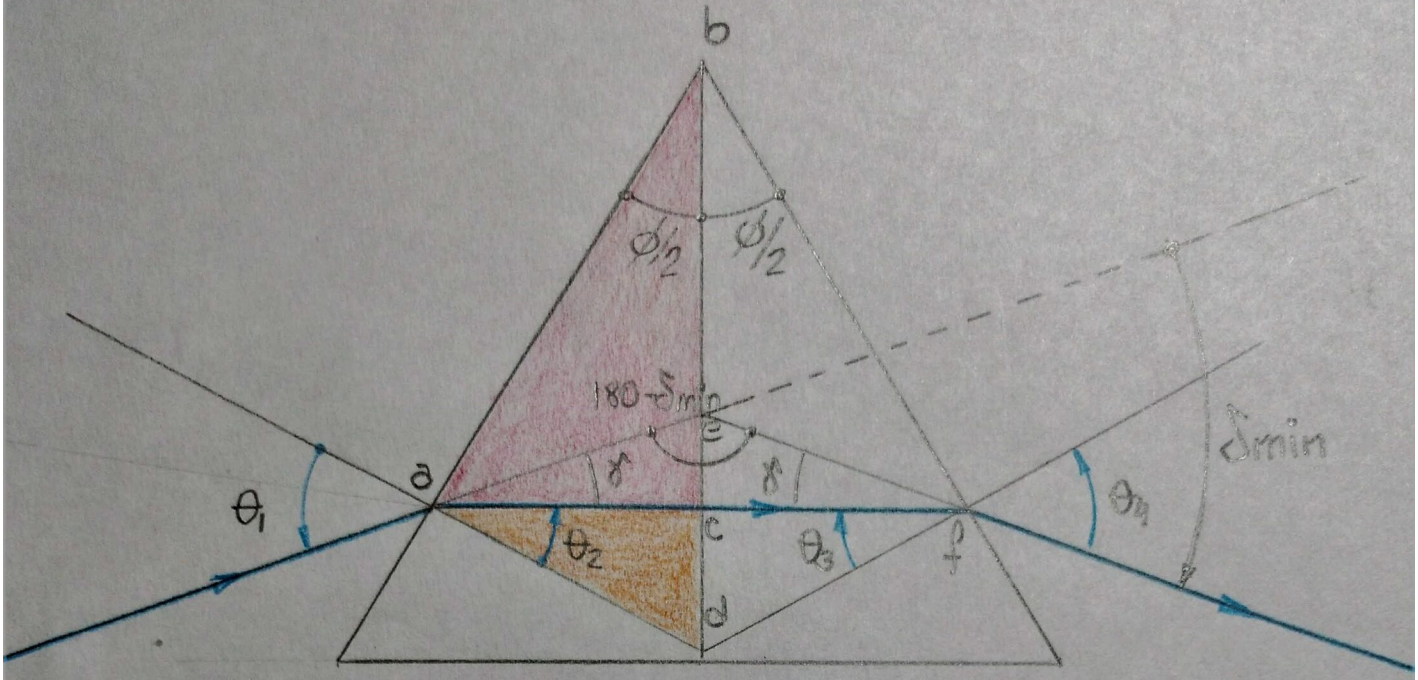
$\boxed{\delta = \theta_1 - \theta_2 + \theta_4 - \theta_3}$

$\delta = \gamma + \gamma'$

$\gamma + \theta_2 = \theta_1$

$\gamma' + \theta_3 = \theta_4$

Repaso del teórico



$\hat{abc}$ semejante $\hat{acd}$ $\Rightarrow$ $\theta_2 = \frac{\phi}{2}$

$\hat{aef}$: $2\gamma + 180 - \delta_{min} = 180$ $\qquad \gamma = \frac{\delta_{min}}{2}$

$$\theta_1 = \theta_2 + \gamma = \frac{\phi + \delta_{min}}{2}$$

$$n_{aire} \cdot \operatorname{sen}\theta_1 = n_v \operatorname{sen}\theta$$

$$\operatorname{sen}\frac{\phi + \delta_{min}}{2} = n_v \operatorname{sen}\frac{\phi}{2}$$

$$\delta_{min} = \theta_1 - \theta_2 + \theta_4 - \theta_3 = 2(\theta_1 - \theta_2)$$

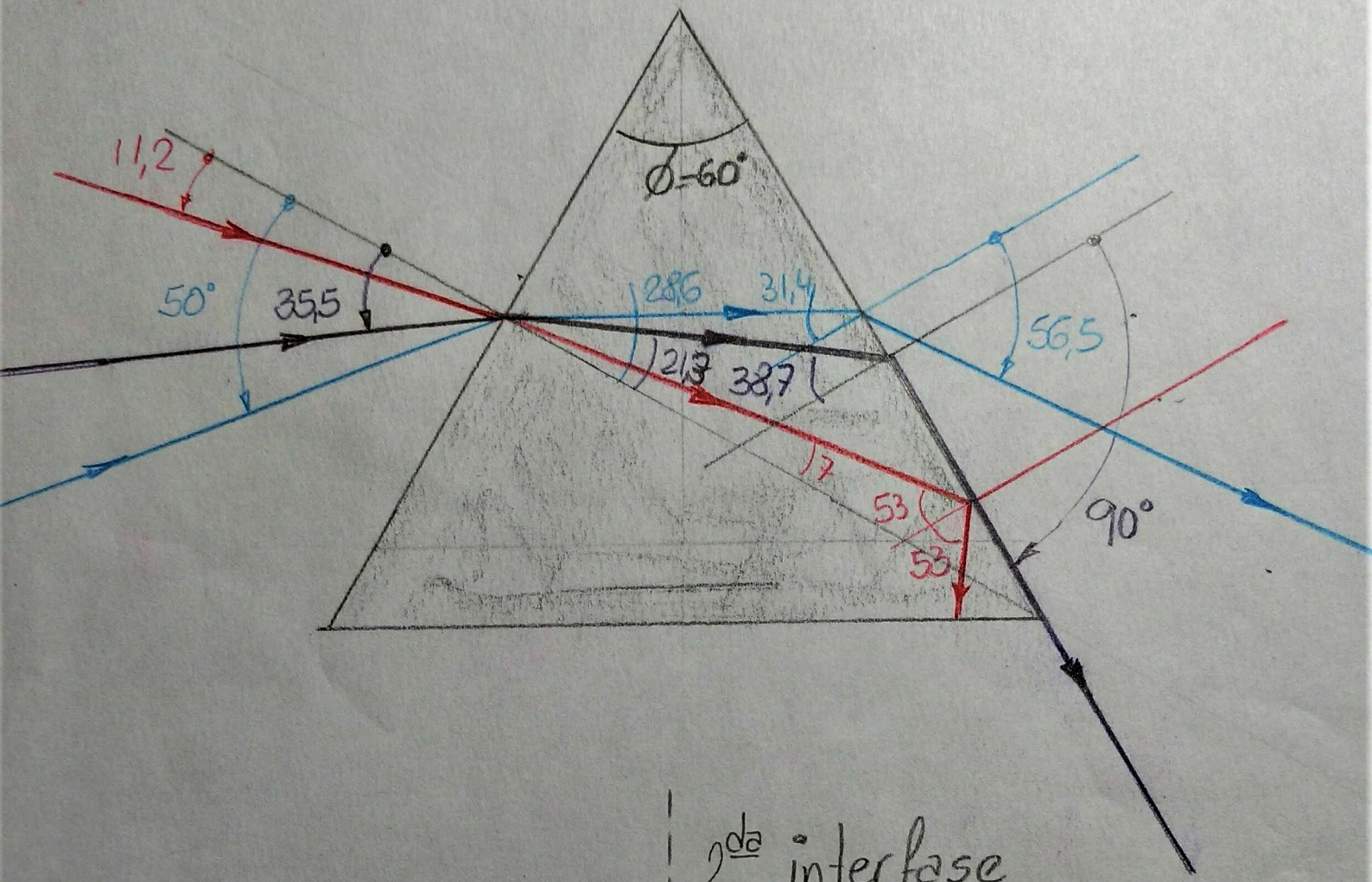
$\delta = \delta_{min} \Rightarrow$ $\theta_1 = \theta_4$

$\theta_3 = \theta_2$

Repaso del teórico

$\phi = 60°$

$n = 1{,}6$



2da interfase

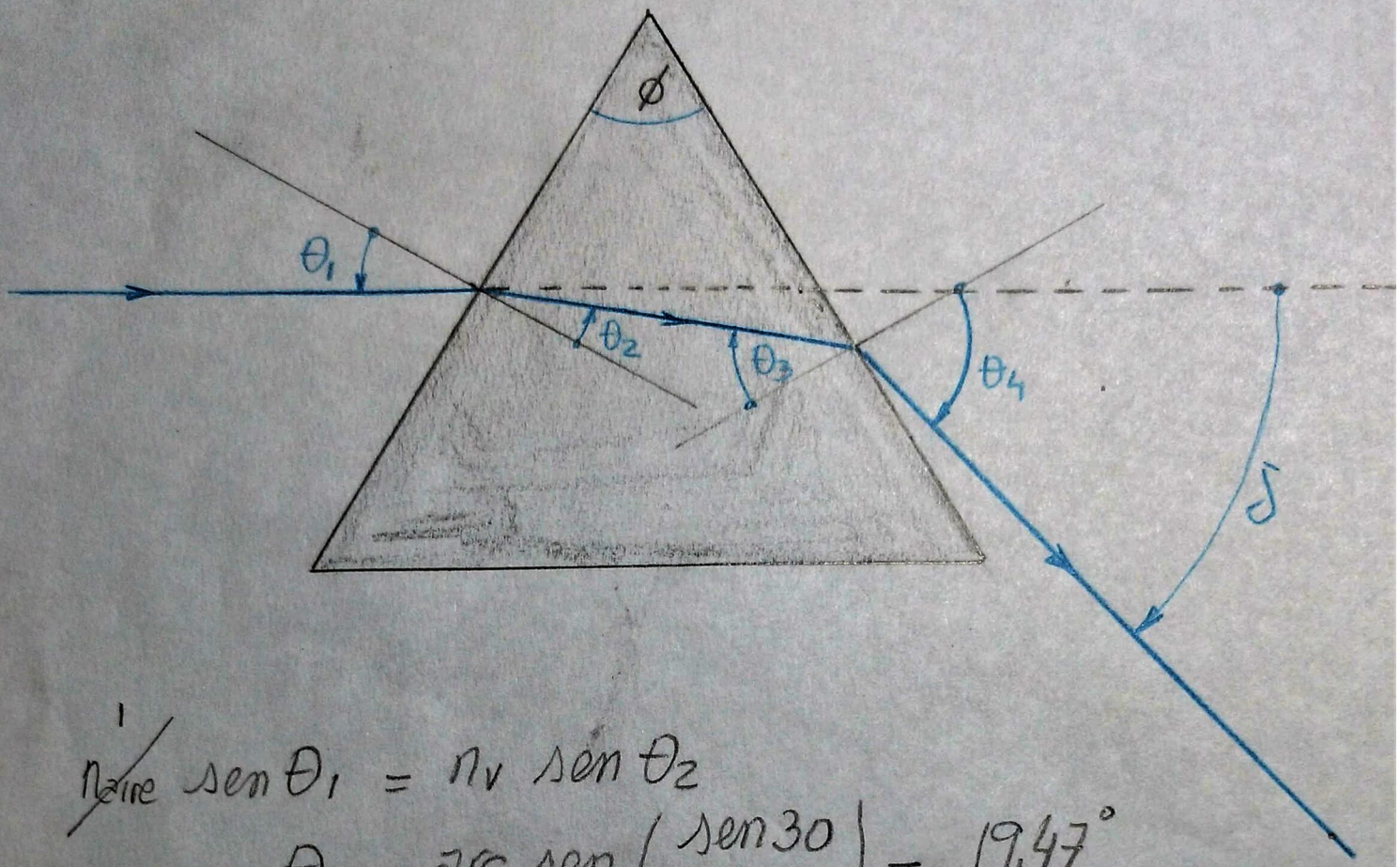
$\theta_1 > 35{,}5$ hay refracción

$\theta_1 < 35{,}5$ hay reflexión interna

21)

$n_V = 1{,}50$

$\phi = 60°$



$n_{ar} \operatorname{sen}\theta_1 = n_V \operatorname{sen}\theta_2$

$$\theta_2 = \operatorname{arc\,sen}\left(\frac{\operatorname{sen} 30}{1{,}50}\right) = 19{,}47°$$

$$\theta_3 = \phi - \theta_2 = 60 - 19.47 = 40{,}53°$$

$n_V \operatorname{sen}\theta_3 = n_{ar} \operatorname{sen}\theta_4$

$$\theta_4 = \operatorname{arc\,sen}(1{,}50 \cdot \operatorname{sen} 40{,}53) = 77{,}10°$$

$$\delta = \theta_1 - \theta_2 + \theta_4 - \theta_3 = 30 - 19{,}47 + 77{,}10 - 40{,}53$$

$$\delta = 42{,}1°$$

$$\delta \neq \delta_{min} \Rightarrow \theta_1 \neq \theta_2$$

22) prisma aquilátero → $\phi = 60°$

$\delta_{min} = 37°$

$n_v = ?$

$$\operatorname{sen}\left(\frac{\phi + \delta_{min}}{2}\right) = n_v \operatorname{sen}\left(\frac{\phi}{2}\right)$$

$$n_v = \frac{\operatorname{sen}\dfrac{\phi + \delta_{min}}{2}}{\operatorname{sen}\dfrac{\phi}{2}}$$

$$n_v = \frac{\operatorname{sen}\dfrac{60+37}{2}}{\operatorname{sen}\dfrac{60}{2}} = 1{,}498$$

23)

$$\operatorname{sen}\frac{\phi+\delta_{min}}{2} = n_v \operatorname{sen}\frac{\phi}{2}$$

$\phi = 15°$ $\qquad n_v = 1{,}52$ (fig 35.15)

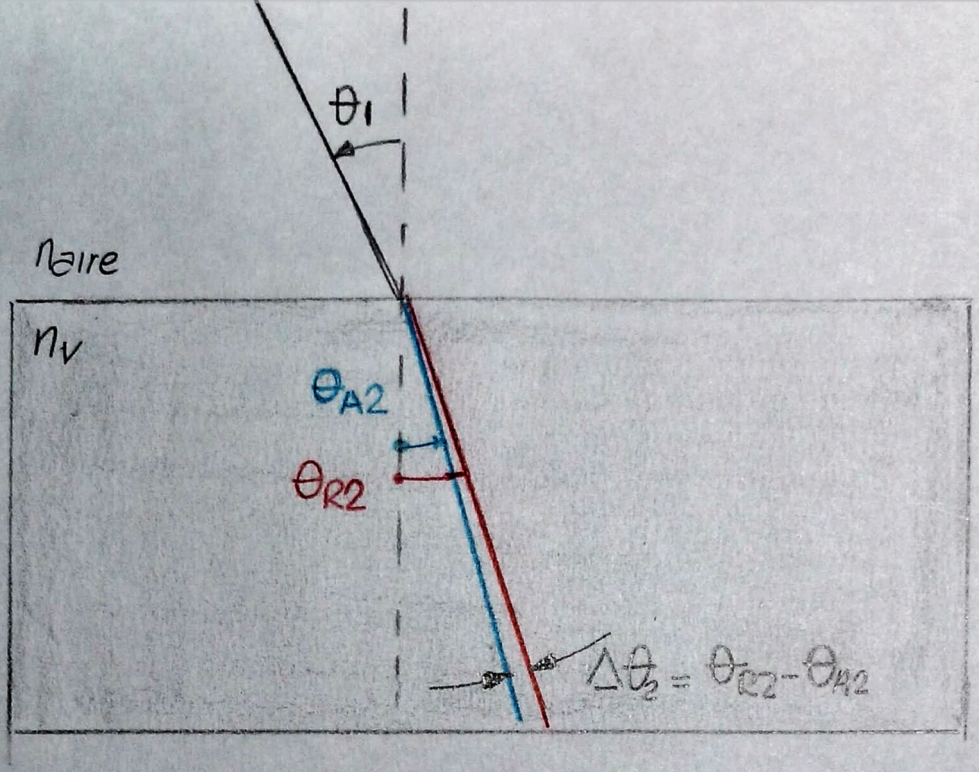
$$\operatorname{sen}\left(\frac{\phi}{2}+\frac{\delta_{min}}{2}\right) = n_v \cdot \operatorname{sen}\frac{\phi}{2}$$

$$\frac{\phi}{2}+\frac{\delta_{min}}{2} = \operatorname{arc\,sen}\left(n_v \cdot \operatorname{sen}\frac{\phi}{2}\right)$$

$$\delta_{min} = 2\left[\operatorname{arc\,sen}\left(n_v \cdot \operatorname{sen}\frac{\phi}{2}\right) - \frac{\phi}{2}\right]$$

$$\delta_{min} = 2\left[\operatorname{arc\,sen}\left(1{,}52 \cdot \operatorname{sen}\frac{15}{2}\right) - \frac{15}{2}\right] = 7{,}89°$$

24)



$n_{VA} = 1{,}6500$ $\quad \lambda_A = 430\,(nm)$

$n_{VR} = 1{,}6150$ $\quad \lambda_R = 680\,(nm)$

$$n_{aire}^{1} \operatorname{sen}\theta_1 = n_{VA} \operatorname{sen}\theta_{A2}$$

$$\theta_{A2} = \operatorname{arc\,sen}\left(\frac{\operatorname{sen} 30}{1{,}65}\right) = 17{,}64°$$

$$n_{aire}^{1} \operatorname{sen}\theta_1 = n_{VR} \operatorname{sen}\theta_{R2}$$

$$\theta_{R2} = \operatorname{arc\,sen}\left(\frac{\operatorname{sen} 30}{1{,}615}\right) = 18{,}04°$$

$$\Delta\theta_2 = \theta_{R2} - \theta_{A2} = 18{,}04 - 17{,}64 = 0{,}4°$$

25) $[\theta] = (rad)$

$\operatorname{sen} 3 = 0,141$ $\quad 3 \neq 0,141$

$\operatorname{sen} 1 = 0,841$ $\quad 1 \neq 0,841$

$\operatorname{sen} 0,5 = 0,479$ $\quad 0,5 \simeq 0,479$

$\operatorname{sen} 0,1 = 0,0998$ $\quad 0,1 = 0,0998$

$0,1\,(rad) = 5,73°$

Un angulo chico ej: $\quad \phi \ll \longrightarrow \delta m \ll$

$$n_v = \frac{\operatorname{sen}\left(\frac{\phi + \delta m}{2}\right)}{\operatorname{sen}\frac{\phi}{2}} = \frac{\frac{\phi + \delta m}{2}}{\frac{\phi}{2}} = \frac{\phi + \delta m}{\phi}$$

$$\phi . n_v = \phi + \delta m$$

$$\delta m = \phi . n_v - \phi = \phi (n_v - 1)$$

forma aprox. $\boxed{\delta m = \phi (n_v - 1)}$ si $[\phi] = (rad)$

26) $n_v = 1{,}62 \qquad \phi = 30° \qquad \phi = 10°$

$$\delta_{min} = 2 \cdot \left[\text{arc sen} \left(1{,}62 \cdot \text{sen} \frac{30°}{2} \right) - \frac{30°}{2} \right] = 19{,}6°$$

$$\delta_{min} = 30 (1{,}62 - 1) = 18{,}6°$$

$$\delta_{min} = 2 \cdot \left[\text{arc sen} \left(1{,}62 \cdot \text{sen} \frac{10°}{2} \right) - \frac{10°}{2} \right] = 6{,}23°$$

$$\delta_{min} = 10 (1{,}62 - 1) = 6{,}20°$$

	30°	10°
ecuación exacta	19,6	6,23
ecuación aproximada	18,6	6,20

27-

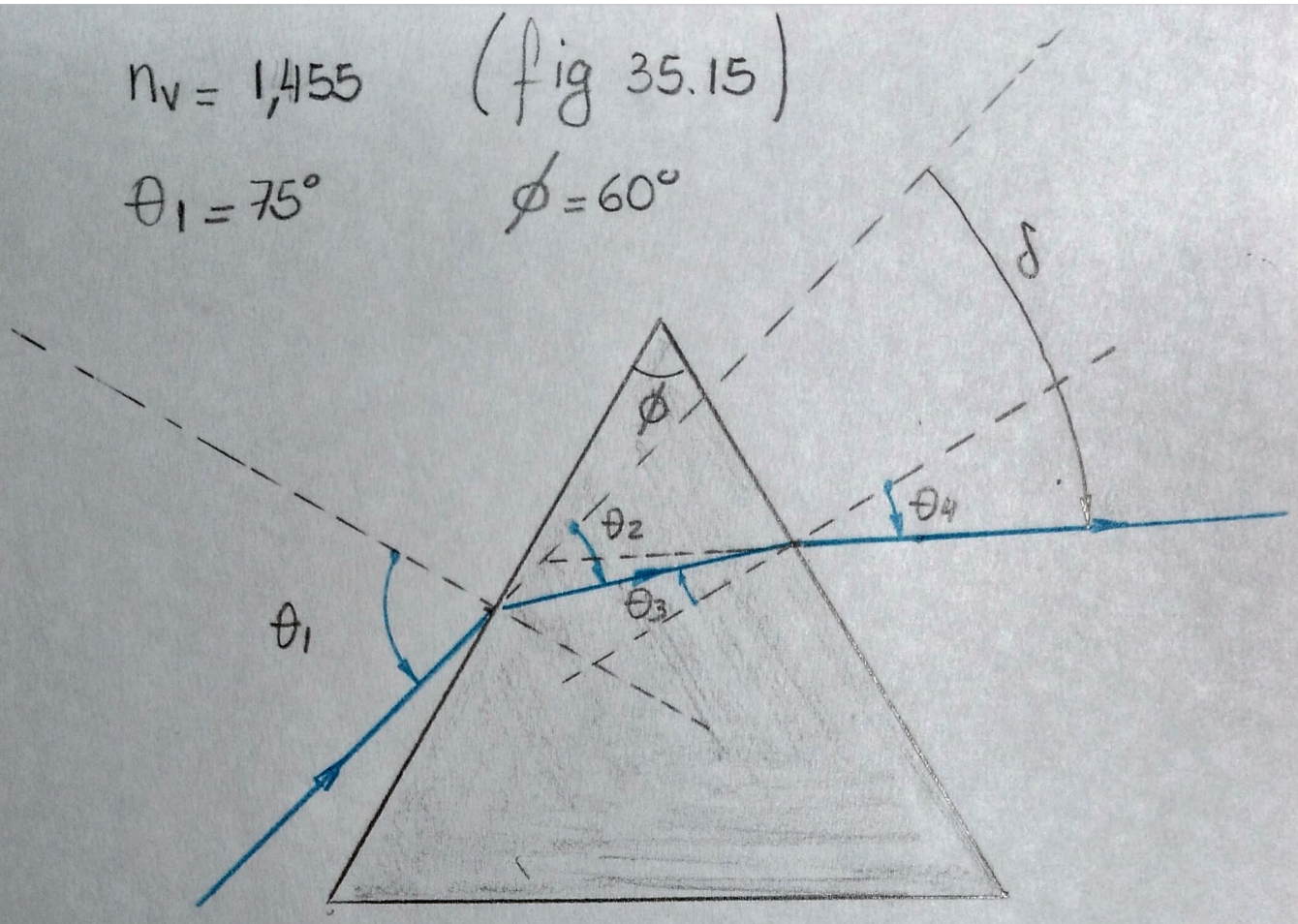
$\delta_{min} = 10°$ $\quad n_v = 1{,}544$ (tabla 35.1)

$$\delta_m = (n_v - 1)\,\phi$$

$$\phi = \frac{\delta_{min}}{n_v - 1} = \frac{10}{1{,}544 - 1} = 18{,}4°$$

28) $n_v = 1{,}455$ (fig 35.15)

$\theta_1 = 75°$ $\phi = 60°$



$n_{ar} \operatorname{sen}\theta_1 = n_v \operatorname{sen}\theta_2$

$\theta_2 = \text{arc sen}\left(\frac{\operatorname{sen} 75}{1{,}455}\right) = 41{,}6°$

$\theta_3 = \phi - \theta_2 = 60 - 41{,}6 = 18{,}4°$

$n_v \operatorname{sen}\theta_3 = n_{ar} \operatorname{sen}\theta_4$

$\theta_4 = \text{arc sen}(1{,}455 \cdot \operatorname{sen} 18{,}4) = 27{,}3°$

$\delta = \theta_1 - \theta_2 + \theta_4 - \theta_3 = 75 - 41{,}6 + 27{,}3 - 18{,}4 = 42{,}3°$

$\delta \neq \delta_{min} \rightarrow \theta_1 \neq \theta_4$

29) $\phi = 50°$ $\quad n_v = n_{CIRCONA\ cúbico} = 2{,}20$

$\delta_{min} = ?$

$$sen\ \frac{\phi + \delta_{min}}{2} = n_v \cdot sen\ \frac{\phi}{2}$$

$$\delta_{min} = 2 \cdot \left[arc\ sen \left(n_v \cdot sen\ \frac{\phi}{2} \right) - \frac{\phi}{2} \right]$$

$$\delta_{min} = 2 \left[arc\ sen \left(2{,}2\ sen\ \frac{50}{2} \right) - \frac{50}{2} \right] = 86{,}8°$$